

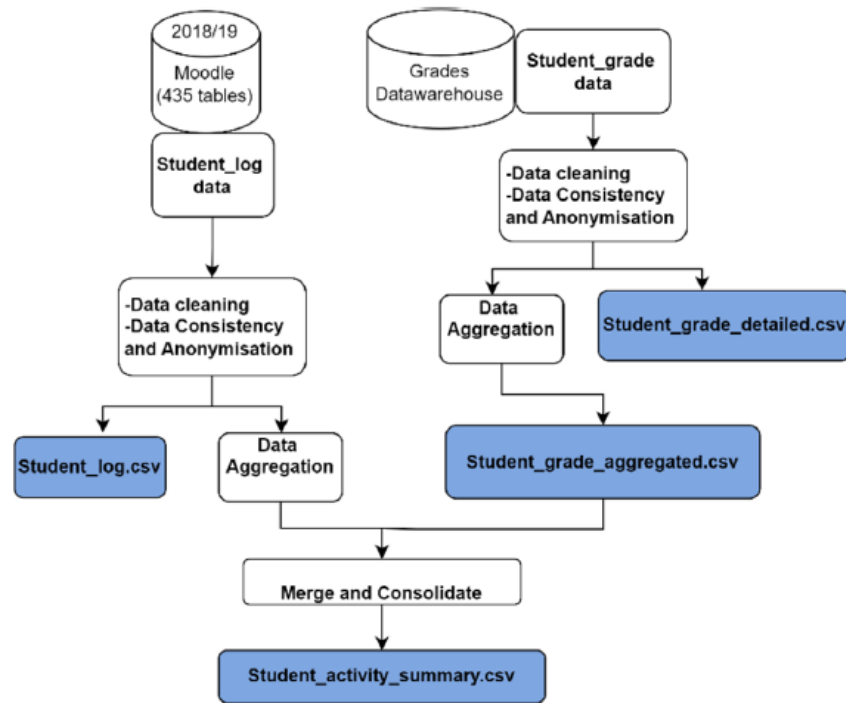
MEMBER TASKS

Role / Section	Member Name	Task Description
Member 1: INTRODUCTION	Corbin Matthews	Introduce topic (application of data analytics/analysis in higher education + goals of the analytics in higher ed); summarize section I. INTRODUCTION (pages 23607 – 23609) in article
Member 2: METHODS	Taghreed Al-Sharafi	Address: "How data is collected, in what format?"; summarize section II. METHODS (pages 23609 – 23610) in article
Member 3: DATA RECORDS & ANALYSIS	Hayden Blue	Summarize section III. DATA RECORDS AND ANALYSIS (pages 23610 – 23614) in article
Member 4: CORRELATION ANALYSIS	Jason Hartz	Summarize section IV. CORRELATION ANALYSIS (pages 23615) in article
Member 5: USAGE NOTES, LIMITATIONS, & CONCLUSION	Aven Pacha	Summarize section V. USAGE NOTES, LIMITATIONS, & CONCLUSION (page 23616) in article
Member 6: EDITOR & COORDINATOR	Vira Vasylevska	Merge segments into 1 video; add transitions if necessary; submit assignment on behalf of group

INTRODUCTION: LEARNING ANALYTICS AND THE NEED FOR SED

- Expansion of distance learning and LMS/VLE usage
- COVID-19 accelerated full reliance on online learning systems
- Learning analytics studies links between engagement, performance, and retention
- Engagement includes assignments, submissions, timing, discussions, and activity patterns
- Limited public datasets exist for structured engagement analysis
- Motivation: Construct the Student Engagement Dataset (SED)

METHODS



Three Phases of Dataset Construction:

- Phase 1: Data Extraction
- Phase 2: Data Cleaning & Anonymization
- Phase 3: Data Aggregation & Feature Engineering

- Data processed using Python (NumPy, Pandas)
- Focused on one academic semester

PHASE 1: DATA EXTRACTION

Extracted raw activity data from Moodle LMS



Used PostgreSQL database queries



Extracted grade data separately as CSV



Focused on engagement-related activities:
(Logins, Assignments, Quizzes, Forums, Attendance)



Restricted to one semester (2018/2019)

PHASE 2: DATA CLEANING & PRIVACY PROTECTION

Removed missing and inconsistent records

Filtered data within semester dates

Removed non-student accounts

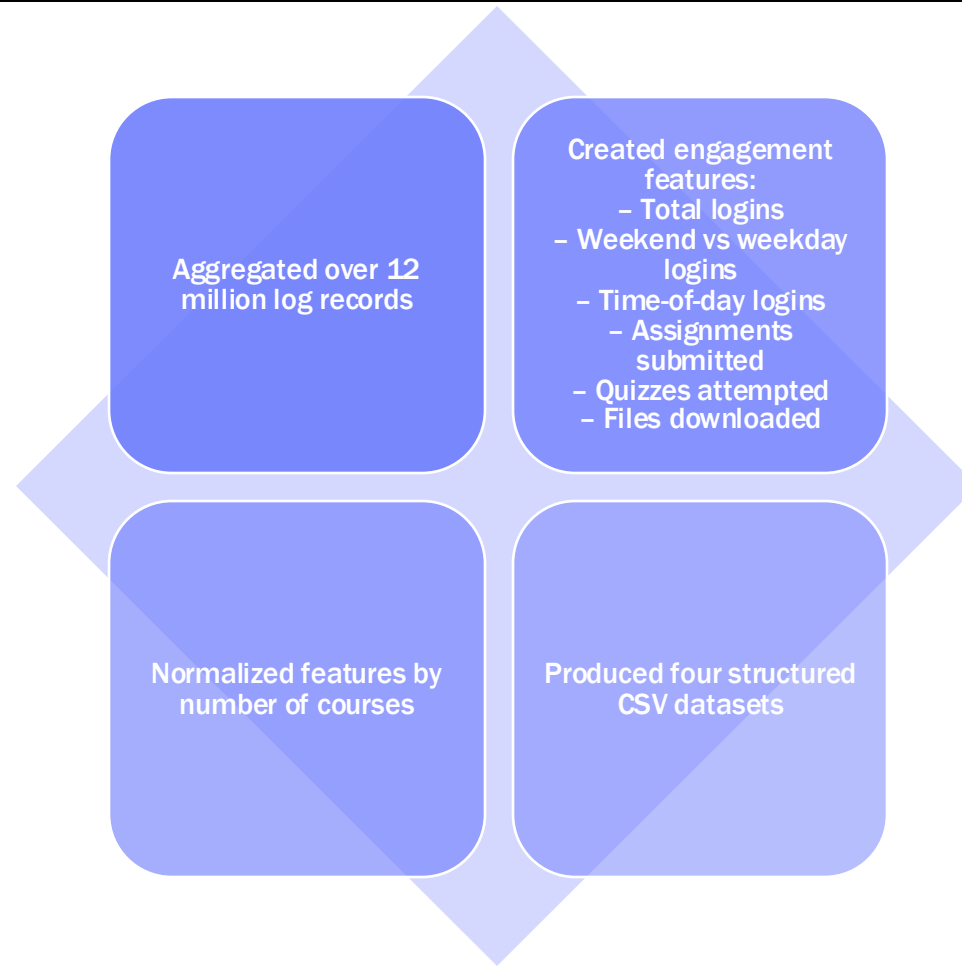
Standardized grading scale

Replaced personal identifiers with:

- Randomized (userid)
- Randomized (courseid)

Ensured compliance with data protection policies

PHASE 3: AGGREGATION AND FEATURE CONSTRUCTION



DATA RECORDS & ANALYSIS

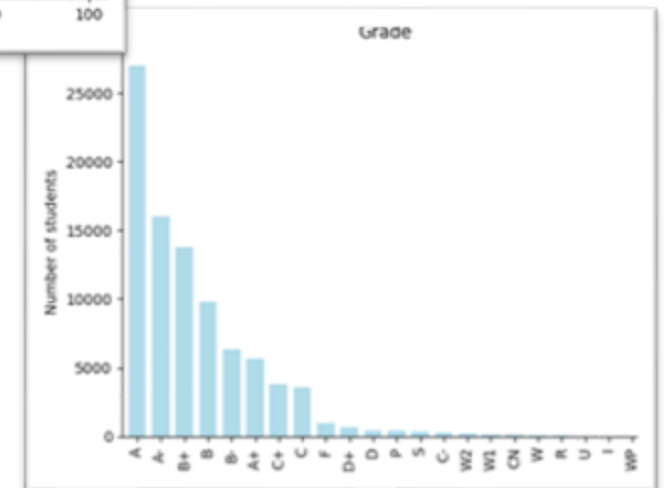
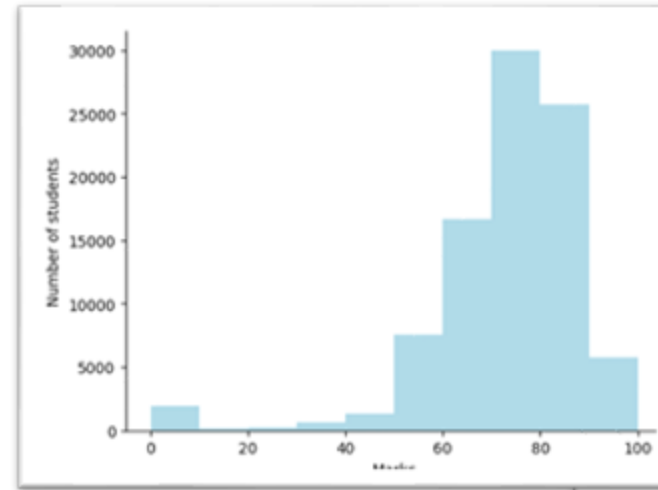
The Role of Data Analytics in Higher Education

DATA ARCHITECTURE OVERVIEW

- **The Connection:** All tables are linked by an anonymized **userid**, where each ID represents a single student.
- **Purpose:** The dataset captures both performance (grades) and learning behavior (LMS activities).
- **The Core Tables:**
 - **Student_grade_detailed.csv:** Granular course-level marks.
 - **Student_grade_aggregated.csv:** High-level student performance summaries.
 - **Student_log.csv:** Massive raw behavioral log data.
 - **Student_activity_summary.csv:** Consolidated analytics with engineered features.

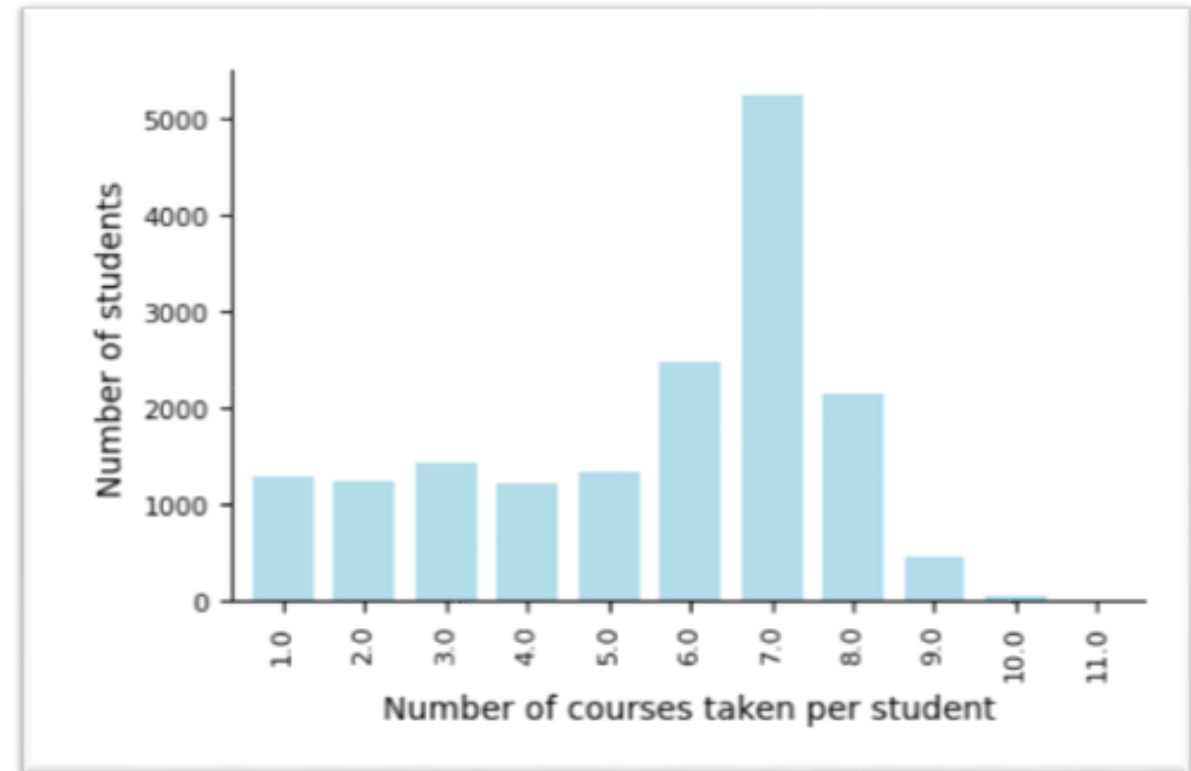
GRADE DISTRIBUTION AND FACULTY INSIGHTS

- **Dataset Scale:** Contains 90,089 rows covering 2,407 unique courses.
- **Key Data Points:** Tracks userid, courseid, formatted agreed mark (0-100), actual grade, and the faculty offering the course.
- **Grading Trends:**
 - **Median Mark:** 75.0
 - **Grade Concentration:** Most students received grades in the A- to A range.
- **Faculty Insight:** "University" courses (common across all faculties) contribute to the high median mark due to relatively lower complexity compared to specialization courses.



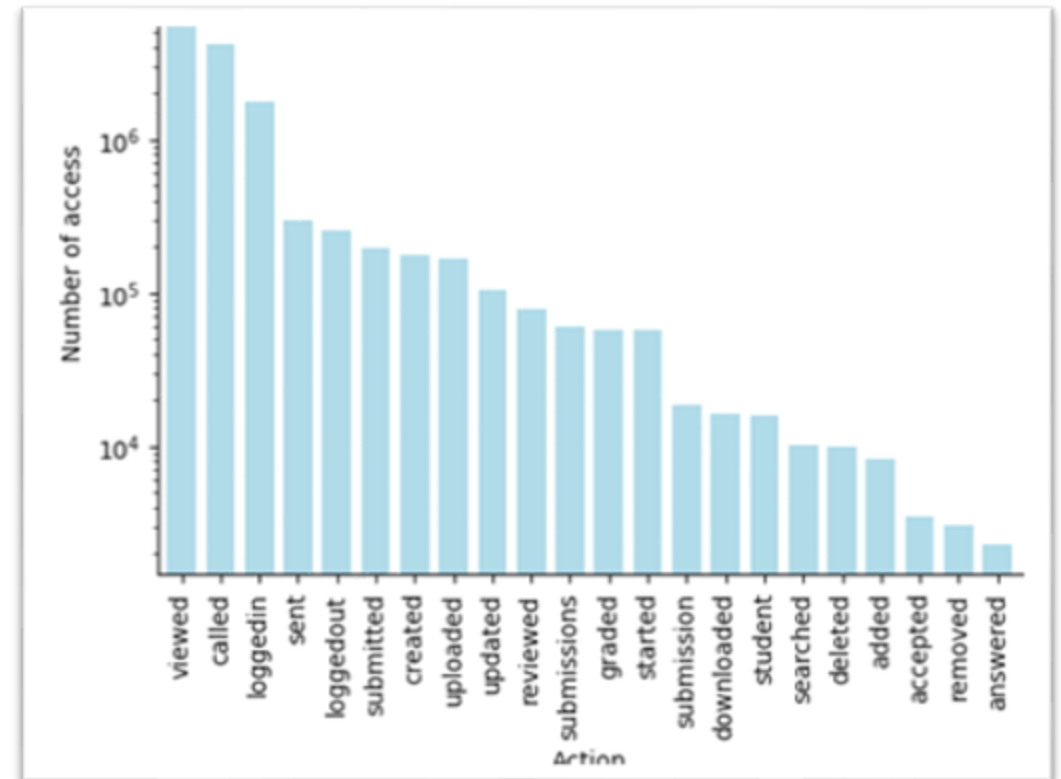
BEHAVIORAL LOGS

- **Dataset:** Student_grade_aggregated.csv
- **Dataset Scale:** Contains 16,909 rows and three columns.
- **Key Metrics:** Tracks userid, number_of_courses completed, and the total marks.
- **Workload Distribution:**
 - **Typical Load:** Most students take 6 to 7 courses across all faculties.
 - **Outliers:** 1,277 students took only one course, while 63 students took ten or more.



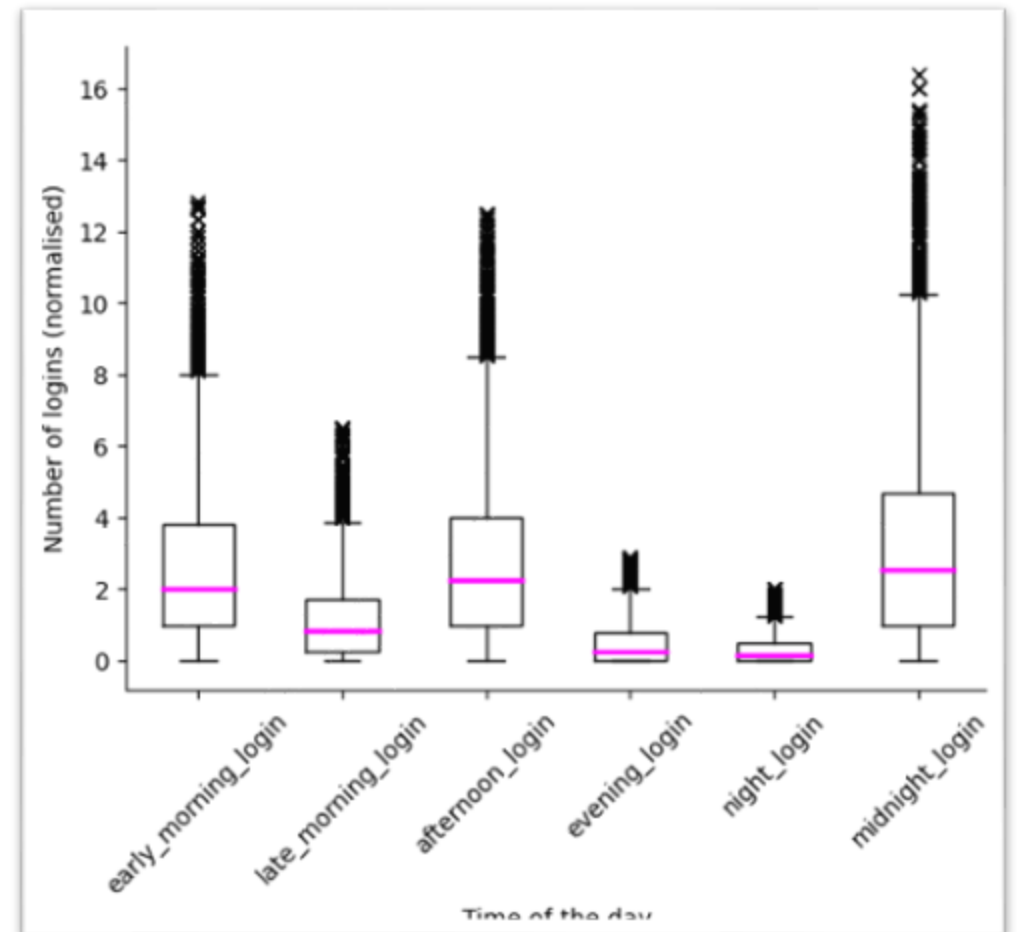
ENGAGEMENT ANALYTICS

- **Dataset:** Student_log.csv
- **Dataset Scale:** The most extensive file with **12,139,424 rows** and six columns.
- **Granularity:** Records specific user interface **components** (36 unique), **actions** (37 unique), and **targets** (74 unique).
- **Key Attribute:** Includes a *timecreated* timestamp for every event.
- **Temporal Insights:** Log data reveals distinct patterns, such as significant drops in student activity during **semester breaks**.



WORKLOAD DISTRIBUTION

- **Dataset:** Student_activity_summary.csv
- **The Synthesis:** Merges aggregated data from Student_log.csv with performance data from Student_grade_aggregated.csv into **16,909** rows and **20** columns.
- **Engineered Features:** Focuses on *when* students study using 4-hour intervals:
- **Time-based:** midnight_login (12 AM–4 AM), early_morning_login (4 AM–8 AM), etc.
- **Resource-based:** Counts for assignments, quizzes, and forum participations.
- **Normalization:** Except for userid and number_of_courses, all columns are **normalized by the number of courses** taken to enable fair comparison between students with different workloads.
- **Peak Activity:** Based on the median, students log in most frequently at **midnight**, followed by the **afternoon** and **early morning**.



CORRELATION ANALYSIS

The Role of Data Analytics in Higher Education

WHY CORRELATION ANALYSIS?

THE RESEARCH SETUP

- Do LMS behaviors (logins, assignments, quizzes) predict student grades?
- Most features are positively skewed — Spearman's rank correlation chosen over Pearson's
- Spearman's does not assume normality, making it the right tool

TWO ANALYSES CONDUCTED

- Case 1: All 16,909 students — full dataset
- Case 2: "Top Students" — averaging ≥ 90 marks ($n = 172$)
- Significance threshold: $\alpha = 0.05$ for both cases

ALL STUDENTS: WHAT THE DATA SHOWS

KEY FINDING (n = 16,909)

- Every engagement feature: $\rho < -0.1$
- Every feature: $P < 0.05$ (statistically significant)
- Weak to negligible negative correlation throughout
- "Strong" threshold is $|\rho| \geq 0.5$ — these results fall far short

WHY SIGNIFICANT, YET WEAK?

- Large sample effect: ~17,000 students makes tiny effects appear significant
- Statistical significance $\neq$ practical significance
- More logins and views do NOT strongly predict better grades

TOP STUDENTS: NARROWING THE FOCUS

Students averaging ≥ 90 marks · $n = 172$ · Only 3 of 18 features reached significance ($\alpha = 0.05$)

Feature	ρ (Spearman)	P-value	Significant?
number_of_courses	-0.2205	0.0036	✓ YES
early_morning_login	-0.1705	0.0252	✓ YES
no_of_assignments	-0.1700	0.0257	✓ YES
total_login	-0.1150	0.1327	✗ No
weekend_login	-0.0925	0.2275	✗ No
no_of_viewed_courses	-0.1248	0.1027	✗ No
no_of_quizzes	-0.1017	0.1840	✗ No

KEY TAKEAWAYS

FINDINGS

- Quality over quantity: top students who logged in less scored higher
- No feature exceeded $|\rho| = 0.22$ — LMS activity is a weak predictor of grades
- Grades reflect prior knowledge, study habits, and course difficulty

IMPLICATIONS FOR DATA ANALYTICS

- Don't confuse statistical significance with practical significance
- Policymakers should require $|\rho| \geq 0.5$ – 0.7 before acting on correlations
- SED enables: at-risk detection, procrastination modeling, course-level analysis

USAGE NOTES, LIMITATIONS & CONCLUSION

The Role of Data Analytics in Higher Education

TOOLS & ANALYTICAL APPROACH

TOOLS USED



Python



NumPy



Pandas



Jupyter Notebook

USAGE NOTES: REALITIES OF EDUCATIONAL DATA

“RAW BUT CLEAN” DATA

- Outliers and zero values are inevitable
- Highly skewed distributions
- Non-normal feature distributions

UNDERUTILIZED LMS FEATURES

- Forum data = only 0.1% of total log data
- Students primarily view courses, download files, submit assignments
- This shows how analytics reveals behavioral patterns, not just final grades.

DEMOGRAPHIC & STRUCTURAL LIMITATIONS

DEMOGRAPHIC LIMITATIONS

- 87.8% Malaysian students
- No gender, age, or socio-economic metadata
- Predictive models may not generalize

STRUCTURAL CONTRIBUTION OF SED

- 16,909 students
- 2,407 courses
- Graduate + postgraduate
- Science + social sciences
- Single semester (consistent demographic profile)

WHY DATA ANALYTICS MATTERS IN HIGHER EDUCATION

THE ROLE OF ANALYTICS

- Understanding patterns of engagement
- Supporting early intervention
- Improving course design
- Enabling evidence-based institutional decisions

